

Differential Equation:

An equation involving independent variables, dependent variables and the derivative of the dependent variable with respect to the independent variable is called a differential equation.

If x is an independent variable, y is a variable dependent on x , then the equation involving

$x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}$ is called differential equation

Examples

- i. $\frac{dy}{dx} = x + \sin x$ ii. $\frac{dy}{dx} = \frac{x^3 - y^3}{xy^2 - x^2y}$
 iii. $x dy + y dx = 0$ iv. $\frac{dy}{dx} + 5y = e^x$
 v. $\frac{d^2y}{dx^2} + 4\left(\frac{dy}{dx}\right)^3 + 3y = 0$ vi. $\log\left(\frac{dy}{dx}\right) + y = 3$
 vii. $\sin\left(\frac{dy}{dx}\right) + y = x$ viii. $y''' + y^2 + e^{y'} = 0$

Order and degree of a Differential Equation

The **order of the highest order derivative** occurring in a differential equation is called the order differential equation.

The **power (index) of the highest order derivative** occurring in a differential equation, after it free from **radicals and fractions**, is called the degree of the differential equation.

Note:

- i) To define degree, the differential equation must be polynomial equation in derivatives (i.e. y', y'', y''' etc)
 ii) Order and degree of the differential equation (if defined) are always positive integer.

Examples

- i. $\left(\frac{dy}{dx}\right)^2 + 5y = \sin x$ It is of order 1 and degree 2.
 ii. $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^3 + 2y = 0$ It is of order 2 and degree 1.
 iii. $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{5/2} = \frac{d^4y}{dx^4}$. It is of order 4 and degree 2
 iv. $\frac{d^4y}{dx^4} + \sin\left(\frac{dy}{dx}\right) = 0$. It is of order 4 and degree cannot be defined due to the term $\sin\left(\frac{dy}{dx}\right)$
 v. $y''' + y^2 + e^{y'} = 0$. It is of order 3 but degree is not defined as it not a polynomial equation in its derivative.

Solution of the differential Equation:

Any relation between the variables, which is free from derivatives and satisfies the given differential equation, is called the solution or primitive of the differential equation.

General Solution

If the solution of a differential equation of n^{th} order contains n arbitrary constants, the solution is called the general solution or the complete primitive or the complete integral of the differential equation.

Examples:

- i) $\frac{dy}{dx} = 4x^3$ is satisfied by $y = x^4 + C$ with arbitrary constants C . So, $y = x^4 + C$ is the general solution of the given differential equation.
 ii) $y = A \sin x + B \cos x$ is the general solution of $\frac{d^2y}{dx^2} + y = 0$
 iii) $y = Ae^x + Be^{-x}$ is the general solution of $\frac{d^2y}{dx^2} - y = 0$

Particular solution:

A solution obtained by giving particular values to arbitrary constants in the general solution is called a particular solution or a particular integral (P.I.).

Examples:

- i) $y = x^4$ is a particular solution of the differential equation $\frac{dy}{dx} = 4x^3$
 ii) $y = 2 \sin x$ is the particular solution of $\frac{d^2y}{dx^2} + y = 0$

Equation of First Order and First Degree:

A differential equation of the form $\frac{dy}{dx} = f(x, y)$ is called first order and first degree differential equation.

Some Standard Forms of Differential Equation:

- i) *Variable Separable Differential Equation*
 ii) *Homogeneous Differential Equation*
 iii) *Exact Differential Equation*
 iv) *Linear Differential Equation*
 v) *Bernoulli's differential Equation*

Variable Separable Differential Equation

A differential equation which can be written in the form $Y dy = X dx$ where, X is a function of x alone and Y is a function of y alone, is called a variable separable differential equation.

Examples:

- i) $x dy + y dx = 0$ ii) $\frac{dy}{dx} = \frac{e^x + 1}{y}$
 iii) $(1 + y^2) dx + (1 + x^2) dy = 0$

Working Rule:

Convert the integral in the form $Y dy = X dx$ then by direct integration, we arrive at its general solution

Homogeneous Differential Equation:

A differential equation of first order and first degree is said to be homogeneous, if it can be written in the

form $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$

Examples:

$$\text{i) } \frac{dy}{dx} = \frac{y}{x} + \sin^2\left(\frac{y}{x}\right) \quad \text{ii) } \frac{dy}{dx} = \frac{x^2+y^2}{2xy}$$

Working Rule:

we put $y = vx$

Then, $\frac{dy}{dx} = v + \frac{dv}{dx}x$ and then the equation becomes,

$$v + x \frac{dv}{dx} = f(v) \quad \text{i.e. } x \frac{dv}{dx} = f(v) - v$$

$$\text{i.e. } \frac{dv}{f(v)-v} = \frac{dx}{x} \text{ (by separating variables)}$$

Integrating both sides and substituting the value of v , we get the required solution.

Exact Differential Equation:

A differential equation of the form

$$Mdx + Ndy = 0 \text{ [i.e. } M(x,y)dx + N(x,y)dy = 0]$$

is said to be exact differential equation if

$$Mdx + Ndy = d[f(x,y)]$$

Where, M and N are function of x or y or both.

Alternatively,

A differential equation $Mdx + Ndy = 0$ will be

$$\text{exact if } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Example:

i) The differential equation $xdy + ydx = 0$ is exact as $xdy + ydx = d(xy)$

ii) The differential equation $xdy - ydx = 0$ is not exact as $xdy - ydx$ cannot be written as $d[f(x,y)]$

However, it can be made exact by multiplying both sides of the equation by $\frac{1}{x^2}$

i.e. the differential equation $\frac{xdy-ydx}{x^2} = 0$ is an exact differential equation since $\frac{xdy-ydx}{x^2} = d\left(\frac{y}{x}\right)$

Here, the factor $\frac{1}{x^2}$ is called an integrating factor(I.F).

Integrating Factor (I.F):

The factor/function which after multiplication convert the given differential equation into exact differential equation is called an integrating factor (I.F).

Example

i) $\frac{1}{x^2}$ is called an integrating factor(I.F) of the differential equation $xdy - ydx = 0$.

Note:

The integrating factor (I.F) is not unique. [i.e. a differential equation may have more than one integrating factors.

Example:

Both the functions $\frac{1}{x^2}$ and $\frac{1}{y^2}$ are the integrating factor (I.F) for the differential equation $xdy - ydx = 0$.

Some formule by inspection:

$$\text{i) } xdy + ydx = d(xy)$$

$$\text{ii) } \frac{xdy-ydx}{x^2} = d\left(\frac{y}{x}\right)$$

$$\text{iii) } \frac{ydx-xdy}{y^2} = d\left(\frac{x}{y}\right)$$

$$\text{iv) } 2xydy + y^2dx = d(xy^2)$$

$$\text{v) } 2xydx + x^2dy = d(x^2y)$$

$$\text{vi) } \frac{2xydy-y^2dx}{x^2} = d\left(\frac{y^2}{x}\right)$$

$$\text{vii) } \frac{2xydx-x^2dy}{y^2} = d\left(\frac{x^2}{y}\right)$$

$$\text{viii) } xdx = d\left(\frac{x^2}{2}\right); ydy = d\left(\frac{y^2}{2}\right)$$

$$\text{ix) } x^2dx = d\left(\frac{x^3}{3}\right); y^2dy = d\left(\frac{y^3}{3}\right)$$

$$\text{x) } \frac{1}{x}dx = d(\ln x); \frac{1}{y}dy = d(\ln y)$$

4. Linear differential Equation:

The differential equation of the form $\frac{dy}{dx} + Py = Q$

Where, P and Q are the functions of x alone or constants (but not of y) is called linear differential equation.

Working Rule:

i) Find $I.F = e^{\int P dx}$

ii) The solution is $y \times (I.F) = \int [Q \times I.F] dx$

Bernoulli's differential Equation:

The differential equation of the form

$$\frac{dy}{dx} + Py = Q \cdot y^n$$

Where, P and Q are the functions of x alone or constants (but not of y) is called Bernoulli's differential equation.

Working Rule:

Q. How do you convert the Bernoulli's differential equation into linear differential equation?

Consider the equation $\frac{dy}{dx} + Py = Q \cdot y^n \dots \dots (i)$

i) Dividing both sides by y^n , we get

$$y^{-n} \frac{dy}{dx} + Py^{1-n} = Q \dots \dots (ii)$$

ii) Put $z = y^{1-n}$ then $\frac{dz}{dx} = (1-n)y^{-n} \cdot \frac{dy}{dx}$

$$\text{i.e. } \frac{1}{(1-n)} \frac{dz}{dx} = y^{-n} \cdot \frac{dy}{dx}$$

So, equation (ii) becomes,

$$\frac{1}{(1-n)} \frac{dz}{dx} + P \cdot z = Q,$$

which is linear differential equation.

Unit: 3 (Differential Equations)

Knowledge Based Questions:

1. Define differential equation. Write one example.
2. Define first order and first degree differential equation with an example.
3. Define order and degree of the differential equation with an example.
4. Write the degree and order of the following differential equations:
 - i. $\left(\frac{dy}{dx}\right)^2 + 5y = \sin x$ (Order 1 and degree 2)
 - ii. $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^3 + 2y = 0$. Order 2 and degree 1.
 - iii. $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{5/2} = \frac{d^4y}{dx^4}$. Order 4 and degree 2
 - iv. $\frac{d^4y}{dx^4} + \sin\left(\frac{dy}{dx}\right) = 0$. It is of order 4 and degree cannot be defined due to the term $\sin\left(\frac{dy}{dx}\right)$
 - v. $y''' + y^2 + e^{y'} = 0$. It is of order 3 but degree is not defined as it not a polynomial equation in its derivative.
5. Define solution of the differential equation.
6. Define general solution of the differential equation.
7. How do you differentiate general solution and particular solution of the differential equation? Explain with suitable example.
8. Define variable separable differential equation. Write its solution.
9. Define homogeneous differential equation an example.
10. Write the working rule for the differential equation of the form $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ Also, write its solution.
11. Define exact differential equation. Give one /one example of the differential equation which is exact and which is not exact.
12. Define integrating factor with an example.
13. If $Mdx + Ndy = 0$ is an exact differential equation then what types of functions are M and N?
14. Write down any three standard form of first order and first degree differential equation and give the example of each.
15. It is given that $y = f(x)$ be a function and $f'(x)$ is its first order derivative in terms of x . Write dy in terms of $f'(x)$ and dx .

16. Consider the differential equation

$$\frac{dy}{dx} + Py = Q.$$

- i) Name the differential equation
 - ii) What does P and Q represent?
 - iii) Write its integrating factor.
 - iv) What is the solution of this differential equation?
17. Is the differential equation $\frac{dx}{dy} + Px = Q$ is linear? If linear, what is its integrating factor and general solution
18. Define Bernoulli's differential equation. At what condition the Bernoulli's differential equation become linear differential equation?

Understanding & Application Based Questions:

19. Show that $y = A\sin x + B\cos x$ is the general solution of $\frac{d^2y}{dx^2} + y = 0$
20. Verify that the function $y = c\sqrt{1-x^2}$ is a solution of the differential equation $\frac{dy}{dx} - \frac{xy}{1+x^2} = 0$
21. Find the differential equation of the family of the curve given by $x^2 + y^2 = 2ax$

Ans: $\frac{dy}{dx} = \frac{y^2 - x^2}{2xy} = 0$
22. Find the differential equation of the family of all straight lines passing through origin.

Ans: $\frac{dy}{dx} = \frac{y}{x}$
23. Find the differential equation of the family of all concentric circles having centre at origin and radius 'r'.

Ans: $\frac{dy}{dx} = -\frac{x}{y}$
24. Find the differential equation of the family of circles touching x-axis at origin.

Ans: $\frac{dy}{dx} = \frac{2xy}{x^2 - y^2}$
25. Find the equation of a curve passing through the point $(-2, 3)$, given that the slope of the tangent to the curve at any point (x, y) is $\frac{2x}{y^2}$

Ans: $y^3 = (3x^2 + 15)$
26. The concept of integration is necessary for solving a differential equation. Justify the statement with an example.
27. What is the difference between general solution and particular solution of the differential equation? Justify.

28. To solve the homogeneous equation of the form $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$, we need to know the concept of variable separable differential equation. Justify

OR

How do you convert homogeneous equation of the form $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ into variable separable differential equation. Justify

OR How do you solve the homogeneous differential equation of the form $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$?

29. The integrating factor (I.F) of the differential equation may not be unique. Are you agree with the statement, justify with suitable example.

30. How do you convert the Bernoulli's differential equation into linear differential equation?

31. Define Bernoulli's differential equation with an example. How do you convert the Bernoulli's differential equation into linear differential equation? Justify with suitable example.

Type: 1 (Variable Separable Differential Equation)

Solve the following differential equation.

- i. $(1 + y^2)dx + (1 + x^2)dy = 0$
- ii. $ydx - xdy = xydy$
- iii. $\sqrt{1 - x^2}dy + \sqrt{1 - y^2}dx = 0$
- iv. $y' = e^{x+y} + 3x^2e^y$
- v. $\frac{dy}{dx} = \frac{xy+y}{xy+x}$
- vi. $\tan x dy + \tan y dx = 0$
- vii. $(1 + x)ydx + (1 + y)x dy = 0$
- viii. $(xy^2 + x)dx + (yx^2 + y)dy = 0$
- ix. $\ln\left(\frac{dy}{dx}\right) = ax + by$
- x. $e^{x-y} dx + e^{y-x} dy = 0$
- xi. $\cos^{-1}\left(\frac{dy}{dx}\right) = x + y$
- xii. $(x + y + 1)^2 dy = dx$

Answers:

- i) $x + y = c(1 - xy)$ ii) $y e^y = cx$
- iii) $x\sqrt{1 - y^2} + y\sqrt{1 - x^2} = c$
- iv) $e^x + e^{-y} + x^3 + c = 0$
- v) $\log \frac{y}{x} + y - x = c$ vi) $\sin x \cdot \sin y = C$
- vii) $\log xy + x + y = C$
- viii) $(x^2 + 1)(y^2 + 1) = C$
- ix) $ae^{-by} + be^{ax} + c$ x) $e^{2x} + e^{2y} = c$
- xi) $\tan \frac{x+y}{2} = x + c$
- xii) xi) $y = \tan^{-1}(x + y + 1) + c$

Type :2 (Homogeneous Differential Equation)

1. Solve the following differential equations.

- i. $x \frac{dy}{dx} = y - x \tan \frac{y}{x}$
- ii. $(x^2 - y^2) \frac{dy}{dx} = xy$
- iii. $x dy - y dx = \sqrt{x^2 + y^2} dx$
- iv. $x \frac{dy}{dx} = y - x \cos^2 \frac{y}{x}$
- v. $\frac{dy}{dx} = \frac{y+1}{x+y+1}$
- vi. $\frac{dy}{dx} = \frac{x-y}{x+y}$, $y = 0$ when $x = 2$
- vii. $(x^3 + y^3)dy - x^2 y dx = 0$
- viii. $x(x - y)dy = y(x + y)dx$
- ix. $\frac{dy}{dx} = \frac{x^2 - 2y^2 + xy}{x^2}$
- x. $\frac{dy}{dx} = \frac{y^3}{x^3}$
- xi) $x \sin \frac{y}{x} dy = [y \sin \frac{y}{x} - x]$

Answers:

- i) $x \sin \frac{y}{x} = c$ ii) $x^2 + 2y^2 \ln cy = 0$
- iii) $y + \sqrt{x^2 + y^2} = cx^2$
- iv) $\tan\left(\frac{y}{x}\right) + \ln x + c$
- v) $y + 1 = ce^{\frac{x}{y+1}}$
- vi) $x^2 - 2xy + y^2 = 4$
- vii) $x^3 = 3y^3 \ln|y| + Cy^3$
- viii) $x + y \log(xy) = 0$
- ix) $\frac{1}{2\sqrt{2}} \ln \left| \frac{x+\sqrt{2}y}{x-\sqrt{2}y} \right| = \ln|x| + C$
- x) $y^2 - x^2 = cx^2 y^2$
- xi) $\cos\left(\frac{y}{x}\right) = \ln x + C$

2) Solve by reducing into homogeneous differential equation

$$\frac{dy}{dx} = \frac{y+2}{y+x+2} \quad \text{Ans: } y + 2 = Ce^{\frac{x}{y+2}}$$

Type 3 (Exact differential Equation)

Solve the following differential equation

- i. $\frac{dy}{dx} = \frac{y-x+1}{y-x+5}$
- ii. $(x^2 + xy^2)dx + (x^2 y + y^2)dy = 0$
- iii. $2xydx - x^2 dy = 0$
- iv. $y(1 + xy)dx - x dy = 0$
- v. $\sec^2 x \cdot \tan y dx + \sec^2 y \tan x dy = 0$
- vi) $\frac{dy}{dx} = \frac{2x+3y+5}{2y-3x+6}$
- vii) $\sin x \cdot \sin y dx - \cos x \cos y dy = 0$

viii) $\frac{dy}{dx} = \frac{x-y}{x+y}$

ix) $(\sin x \cdot \tan y - 1)dx - \cos x \cdot \sec^2 y dy = 0$

x) $(e^y + 1) \cos x dx + e^x \sin x dy = 0$

xi) $\sec^2 y (1 + x^2)dy + 2x \tan y dx = 0$

Answers:

i) $x^2 + y^2 - 2xy + 10y - 2x = c$

ii) $2x^3 + 2y^3 + 3x^2y^2 = c$

iii) $x^2 = cy$

iv) $\frac{x}{y} + \frac{x^2}{2} = c$

v) $\tan x \cdot \tan y = c$

vi) $x^2 - y^2 + 3xy + 5x - 6y = c$

vii) $\cos x \cdot \sin y = C$

viii) $-x^2 + y^2 + 2xy = C$

ix) $\cos x \cdot \tan y + x = C$

x) $(e^y + 1) \sin x = C$

xi) $(1 + x^2) \tan y = C$

Type: 4 (Linear Differential Equation)

Solve the following differential equation

i. $\cos^2 x \frac{dy}{dx} + y = 1$

ii. $\sin x \frac{dy}{dx} + \cos x \cdot y = x \sin x$

iii. $(1 + x^2) \frac{dy}{dx} + y = e^{\tan^{-1} x}$

iv. $(1 + x) \frac{dy}{dx} - xy = 1 - x$

v. $\frac{dy}{dx} + \frac{2x}{1+x^2} y = \frac{1}{(1+x^2)^2}$

vi. $\frac{dy}{dx} + 2 \cdot \tan x \cdot y = \sin x$

vii. $\frac{dy}{dx} + \frac{y}{x^2} = \frac{1}{x^2}$

viii) $\sin 2x \frac{dy}{dx} - y = \tan x$

ix) $x \frac{dy}{dx} + 2y = x \cos x$

x. $(1 - x^2) \frac{dy}{dx} - xy = 1$

xi) $x \cdot \frac{dy}{dx} + 2y = x^2 \ln(x)$

xii. $\frac{dy}{dx} + \frac{y}{x} = \sin x^2$

xiii) $x \cdot \ln x \cdot \frac{dy}{dx} + y = \frac{2}{x} \ln(x)$

Answers:

i) $y = 1 + ce^{\tan^{-1} x}$

ii) $y \sin x + x \cos x = \sin x + c$

iii) $y = \frac{1}{2} e^{\tan^{-1} x} + ce^{-\tan^{-1} x}$

iv) $(1 + x)y = x + ce^x$

v) $(1 + x^2)y = \tan^{-1} x + c$

vi) $y = \cos x + C \cos^2 x$ vii) $y = 1 + Ce^{\frac{1}{x}}$

viii) $y = \tan x + c\sqrt{\tan x}$

ix) $x^2 y = x^2 \sin x + 2x \cos x - 2 \sin x + c$

x) $y\sqrt{1-x^2} = \sin^{-1} x + c$

xi) $y = \frac{1}{4} x^2 \ln(x) - \frac{1}{16} x^2 + c x^{-2}$

xii) $xy = -\frac{1}{2} \cos x^2 + C$

xiii) $y \ln(x) = -\frac{2}{x} (1 + \ln(x)) + C$

Type: 5 (Bernoulli's differential equation)

i. $\frac{dy}{dx} + y = xy^2$

ii. $\frac{dy}{dx} + y \tan x = y^3 \sec x$

iii. $\frac{dy}{dx} + \frac{1}{x} \cdot y = x^2 y^6$

Answers

i) $xy + y + cye^x = 1$

ii) $(C - 2 \sin x)y^2 = \cos^2 x$

iii) $y^5(5x^3 + Cx^5) = 2$

Other Extra Questions:

1) An exact differential equation have always the mathematical form like $M(x, y)dx + N(x, y)dy = 0$. Take simple example to justify the statement with the characteristics of $M(x, y)$ and $N(x, y)$.

2) A differential equation of first order and first degree is homogeneous if it satisfies the condition $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$. Justify the statement with an example and solve it.

3) Study the differential equation and answer the following equations

$$(1 + x^2) \frac{dy}{dx} + k \cdot y = e^{\tan^{-1} x}$$

- Write the order of the differential equation
- Write the degree of the differential equation
- Write the type of the differential equation
- If k is not a constant number, then whose function may be the k ?
- Write any one use of differential equation

4) Reduce the differential equation $\frac{dy}{dx} + y = xy^2$ into linear form and solve it.

Ans: $xy + y + cy e^x + 1$